United States Patent	12397952
Kind Code	B2
Date of Patent	August 26, 2025
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Device and method for opening plastic bag

Abstract

A plastic bag opening device, which opens an inlet of a plastic bag having a pocket shape, includes an elevating member that lifts the plastic bag in a state in which a first surface of the plastic bag is adhered; and an air blower that injects air toward the inlet of the plastic bag adhered to the elevating member so that the inlet of the plastic bag is opened. Accordingly, the inlet of the plastic bag may be completely opened.

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Appl. No.:	**18/563313**
Filed (or PCT Filed):	**November 03, 2022**
PCT No.:	**PCT/KR2022/017132**
PCT Pub. No.:	**WO2023/113229**
PCT Pub. Date:	June 22, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240270424 A1	Aug. 15, 2024

Foreign Application Priority Data

KR	10-2021-0181100	Dec. 16, 2021

Publication Classification

Int. Cl.: **B65B43/28** (20060101); **B65B43/30** (20060101); **B65B43/36** (20060101)

U.S. Cl.:

CPC **B65B43/28** (20130101); **B65B43/30** (20130101); **B65B43/36** (20130101);

Field of Classification Search

CPC: B65B (43/28); B65B (43/18); B65B (43/34); B65B (43/30); B65B (43/36); B31B (70/003)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10836525	12/2019	Alspaugh	N/A	N/A
2006/0207225	12/2005	Fukuda	N/A	N/A
2008/0110132	12/2007	Yan Ruz	53/385.1	B65B 43/30

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2527693	12/1976	DE	B65B 43/36
57-110010	12/1981	JP	N/A
59-1928	12/1983	JP	N/A
64-20409	12/1988	JP	N/A
5-254517	12/1992	JP	N/A
8-156917	12/1995	JP	N/A
9-226724	12/1996	JP	N/A
4918152	12/2011	JP	N/A
2013-142001	12/2012	JP	N/A
2018-135119	12/2017	JP	N/A
2020-172310	12/2019	JP	N/A
10-0391971	12/2002	KR	N/A
10-0438122	12/2003	KR	N/A
10-0460965	12/2003	KR	N/A
20-0432422	12/2005	KR	N/A
10-2012-0026315	12/2011	KR	N/A
10-1242769	12/2012	KR	N/A
10-1281542	12/2012	KR	N/A
10-1446670	12/2013	KR	N/A
10-2018-0005372	12/2017	KR	N/A
10-2018-0024372	12/2017	KR	N/A
10-1841425	12/2017	KR	N/A
10-2019-0103842	12/2018	KR	N/A
10-2092123	12/2019	KR	N/A

OTHER PUBLICATIONS

Translation of DE 2527693 A1 (Year: 1977). cited by examiner
Translation of JP 08156917 (Year: 1994). cited by examiner
Partial European Search Report for European Application No. 22907700.3, dated Sep. 18, 2024. cited by applicant
International Search Report issued in PCT/KR2022/017132 (PCT/ISA/210), mailed on Feb. 15, 2023. cited by applicant

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Background/Summary

TECHNICAL FIELD
Cross-Reference to Related Applications
(1) The present application claims the benefit of the priority of Korean Patent Application No. 10-2021-0181100, filed on Dec. 16, 2021, which is hereby incorporated by reference in its entirety.
Technical Field
(2) The present disclosure relates to a device and method for opening a plastic bag, which is capable of automatically opening an inlet of the plastic bag.
BACKGROUND ART
(3) In general, products such as foods, medicines and articles, are packed and distributed in packaging bags in the form of a bag, and a synthetic resin is generally used as a material of the packaging bags.
(4) For example, the packaging bags may be provided as plastic bags in the form of an envelope, and the plastic bags each have a structure in which surfaces other than one surface are sealed. That is, the non-sealed one surface of the packaging bag serves as an inlet for accommodating the products.
(5) As the industry develops, a plastic bag opening device for accommodating many products in plastic bags in a short time is developed. The plastic bag opening device includes first and second adherence pads that adhere to two opposite surfaces of the plastic bag, respectively, and a movable member that moves the first and second adherence pads in directions that are away from each other, to open the inlet of the plastic bag.
(6) However, there is a limit to how the plastic bag opening device according to the related art completely opens the inlet of the plastic bag solely by the first and second adherence pads. That is, the inlet of the plastic bag is difficult to open because of frictional force and static electricity. Accordingly, capacity utilization may be reduced and packaging failure may occur because of a plastic bag opening error.
DISCLOSURE OF THE INVENTION
Technical Problem
(7) An object of the present invention for solving the above problems is to provide a device for opening a plastic bag, which includes an air blower to more easily open an inlet of the plastic bag, and particularly, to completely open the inlet of the plastic bag and accordingly, may increase capacity utilization and prevent an occurrence of defective packaging, and a method for opening a plastic bag using the device.
Technical Solution

(8) An object of the present invention for solving the above problems is to provide a plastic bag opening device that opens an inlet of a plastic bag having a pocket shape. The plastic bag opening device may include: a lifter that lifts the plastic bag in a state in which a first surface of the plastic bag is adhered; and an air blower that injects air toward the inlet of the plastic bag adhered to the lifter so that the inlet of the plastic bag is opened.
(9) The lifter may include an adherence pad, which adheres to the first surface of the plastic bag, and an elevating part that causes the adherence pad to ascend so as to lift the plastic bag adhered to the adherence pad up to an imaginary horizontal line at which the air blower is disposed.
(10) The air blower may include an injection nozzle, which injects the air toward the inlet of the plastic bag so that the inlet of the plastic bag is opened, the injection nozzle may have an air injection opening inclinable downward at a set angle with respect to the imaginary horizontal line; and a frame to which the injection nozzle is coupled.
(11) The set angle may be set to 3° to 10°.
(12) The injection nozzle may be coupled to the frame so that an angle of the injection nozzle with respect to the frame is adjustable up and down or left and right.
(13) The plastic bag opening device may further include a rotator that rotates the lifter so that the inlet of the plastic bag adhered to the lifter faces upward.
(14) The plastic bag opening device may further include a pair of fingers that are inserted into the inlet of the plastic bag adhered to the lifter to pull and expand the inlet of the plastic bag in opposite directions.
(15) The plastic bag opening device may further include a movable member that moves the lifter away from the pair of fingers such that the inlet of the plastic bag is expanded between the pair of fingers and the lifter.
(16) The present invention relates to a plastic bag opening method that opens an inlet of a plastic bag having a pocket shape. The plastic bag opening method may include: an elevating process of adhering a first surface of the plastic bag to a lifter and then lifting the plastic bag using the lifter; and an opening process of opening the inlet of the plastic bag using an air blower to inject air toward the inlet of the plastic bag adhered to the lifter so that the air injected from the air blower flows into the plastic bag through the inlet.
(17) In the elevating process, the lifter may lift the plastic bag up to an imaginary horizontal line at which the air blower is disposed. In the opening process, the air blower may inject the air toward the inlet of the plastic bag so that the air is injected downward at a set angle with respect to the imaginary horizontal line.
(18) The set angle may be set to 3° to 10°.
(19) The plastic bag opening method may include, after the opening process, a rotating process of rotating the lifter using a rotator so that the inlet of the plastic bag adhered to the lifter is disposed to face upward.
(20) The plastic bag opening method may include, after the rotation process, a primary expanding process of pulling and expanding the inlet of the plastic bag in opposite directions by inserting a pair of fingers into the inlet of the plastic bag adhered to the lifter and moving the pair of fingers in directions that are away from each other.
(21) The plastic bag opening method may include, after the primary expanding process, a secondary expanding process of expanding the inlet of the plastic bag between the pair of fingers and the lifter using a movable member to move the lifter away from the pair of fingers.
Advantageous Effects
(22) The plastic bag opening device according to the present invention may include the air blower to inject the air into the inlet of the plastic bag and completely open the inlet of the plastic bag. Accordingly, the capacity utilization may increase, and the occurrence of the defective packaging may be prevented.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a perspective view illustrating a plastic bag according to the present invention.
(2) FIG. **2** is a side view illustrating a plastic bag opening device according to a first embodiment of the present invention.
(3) FIG. **3** is a perspective view illustrating an air blower of the plastic bag opening device according to the first embodiment of the present invention.
(4) FIG. **4** is a side view illustrating a rotating member of the plastic bag opening device according to the first embodiment of the present invention.
(5) FIG. **5** is a perspective illustrating a finger member of the plastic bag opening device according to the first embodiment of the present invention.
(6) FIG. **6** is a plan view illustrating an operation state of a finger member of the plastic bag opening device according to the first embodiment of the present invention.
(7) FIG. **7** is a plan view illustrating a movable member of the plastic bag opening device according to the first embodiment of the present invention.
(8) FIG. **8** is a flowchart illustrating a plastic bag opening method according to a first embodiment of the present invention.
(9) FIG. **9** is a perspective illustrating a plastic bag opening device according to a second embodiment of the present invention.
(10) FIG. **10** is a cross-sectional view taken along a line A-A in FIG. **9**.
MODE FOR CARRYING OUT THE INVENTION
(11) Hereinafter, embodiments of the present invention will be described in detail with reference to the accompanying drawings to enable those skilled in the art to which the present invention pertains to easily carry out the present invention. The present invention may, however, be embodied in different forms and should not be construed as limited by the embodiments set forth herein. The parts unrelated to the description will be ruled out in the drawings in order to clearly describe the present invention. Like reference numerals refer to like elements throughout the whole specification.
(12) Referring to FIG. **1**, a plastic bag **10** has a structure in the form of a pocket having an end in which an inlet **11** is formed. The plastic bag **10** having such a structure is kept in a state in which the plastic bag **10** is pressed flat so that the first surface **12** and a second surface **13** are in close contact with each other. When using the plastic bag **10**, the inlet **11** thereof is opened to store or pack products, etc.
(13) However, there is a problem that the plastic bag **10** is not easily opened because of frictional force or static electricity occurring between the first surface **12** and the second surface **13**.
(14) A plastic bag opening device **1** according to a first embodiment of the present invention may automatically open the inlet **11** of the plastic bag **10**, and particularly, completely open the inlet of the plastic bag. Accordingly, efficiency of accommodating articles may be improved.
(15) Hereinafter, the plastic bag opening device according to the first embodiment of the present invention will be described in detail with reference to the drawings.
(16) [Plastic Bag Opening Device According to First Embodiment of Present Invention]
(17) As illustrated in FIGS. **1** to **3**, the plastic bag opening device **1** according to the first embodiment of the present invention has a structure that opens the inlet **11** of the plastic bag **10** having a pocket shape. That is, the plastic bag opening device **1** according to the first embodiment of the present invention includes an elevating member **100**, which lifts the plastic bag **10** in a state in which a first surface **12** (a top surface of the plastic bag in FIG. **1**) of the plastic bag **10** is adhered, and an air blower **200** which injects air toward the inlet **11** of the plastic bag **10** adhered to the elevating member **100** so that the inlet **11** of the plastic bag **10** is opened to be widened.

(18) Elevating Member
(19) The elevating member **100** is intended to lift the plastic bag **10**.
(20) That is, as illustrated in FIG. **2**, the elevating member **100** includes an adherence pad **110**, which adheres the first surface **12** of the plastic bag **10**, and an elevating part **120** which allows the adherence pad **110** to ascend and lifts the plastic bag **10** adhered to the adherence pad **110** up to a horizontal line (O) at which the air blower **200** is disposed.
(21) Here, the adherence pad **110** may be provided in plurality so as to lift the entire first surface **12** provided at a side of the inlet **11** of the plastic bag **10**, and preferably, five or more adherence pads **110** may be provided.
(22) In particular, the adherence pad **110** may have a structure in which an air suction force is used to adhere to the first surface **12** of the plastic bag **10**. The elevating part **120** may be provided as a hydraulic cylinder or pneumatic cylinder that allows the adherence pad **110** to ascend or return to an initial position.
(23) The elevating member **100** having the structure as described above may stably adhere to the first surface **12** of the plastic bag **10** and effectively lift the adhered plastic bag **10**.
(24) Air Blower
(25) The air blower **200** is intended to open the inlet of the plastic bag.
(26) That is, as illustrated in FIG. **3**, the air blower **200** includes an injection nozzle **210**, which injects the air toward the inlet **11** of the plastic bag **10** so that the inlet **11** of the plastic bag **10** is opened to be widened, and a frame **220** to which the injection nozzle **210** is coupled.
(27) Here, the frame **220** is formed to extend in a direction corresponding to the inlet **11** of the plastic bag **10** (a front and rear direction in FIG. **1**), and the injection nozzle **210** is provided in plurality at set intervals on one surface of the frame **220**, which faces the plastic bag **10**.
(28) Here, the injection nozzle **210** may be provided to face the inlet **11** of the plastic bag **10** so that an air injection opening through which the air is injected is inclined downward at a set angle α with respect to the horizontal line (O). The air injected from the injection nozzle **210** may be injected toward the second surface **13** rather than the first surface **12** of the plastic bag **10**. Accordingly, the second surface **13** of the plastic bag **10**, which adheres to the first surface **12** by static electricity, etc., may be easily separated to open the inlet **11** of the plastic bag **10**.
(29) Here, the set angle α may be set to 3° to 10°, and preferably, may be provided as 5° to 8°. When an injection angle of the injection nozzle **210** is 3° or less, there is no significant difference compared to the case of injecting toward the horizontal line. When the injection angle is 10° or greater, the air may be injected outside the inlet **11**, and thus the efficiency may be reduced.
(30) In an operation state of the plastic bag opening device **1** according to the first embodiment of the present invention having the structure as described above, the elevating member **100** lifts the first surface **12** of the plastic bag **10** in the adhered state. Here, the second surface (a bottom surface of the plastic bag in FIG. **1**) of the plastic bag **10** may have a portion adhering to the first surface by static electricity. The entirety of the second surface **13** may also adhere to the first surface, and the entirety of the second surface **13** may also be separated from the first surface due to the load.
(31) Then, the air blower **200** is used to inject the air toward the inlet **11** of the plastic bag **10**. The air injected by the air blower **200** may flow into the inlet **11** to open the inlet **11**, and part of the air having flowed into the plastic bag **10** may be discharged through the inlet **11** to reopen the inlet **11**.
(32) Thus, as the plastic bag opening device **1** according to the first embodiment of the present invention includes the elevating member **100** and the air blower **200**, the inlet **11** of the plastic bag **10** may be opened to be completely widened.
(33) Rotating Member
(34) The plastic bag opening device **1** according to the first embodiment of the present invention may further include a rotating member **400** that rotates the elevating member **100** so that the inlet **11** of the plastic bag **10** faces upward.
(35) That is, as the rotating member **400** allows the inlet **11** of the plastic bag **10** to be disposed to

face upward as illustrated in FIG. **4**, a product may be more easily introduced into the plastic bag **10**.
(36) In one example, the rotating member **400** includes a rotating body **410**, to which the elevating member **100** is rotatably coupled, and a rotating part **420** which rotates the elevating member **100** coupled to the rotating body **310** so that the inlet **11** of the plastic bag **10** adhered to the elevating member **100** is disposed to face upward. The rotating part **420** may be a driving motor.
(37) Finger Member
(38) The plastic bag opening device **1** according to the first embodiment of the present invention may further include a finger member **300** that pulls the inlet **11** of the plastic bag **10** to primarily expand the inlet **11** of the plastic bag **10** in two opposite directions.
(39) That is, as illustrated in FIGS. **5** and **6**, the finger member **300** includes a pair of fingers **310**, and the pair of fingers **310** pulls the inlet **11** of the plastic bag **10** adhered to the elevating member **100** in the two opposite directions while moving in directions that are away from each other, in a state inserted into the inlet **11** of the plastic bag **10**.
(40) In one example, the finger member **300** includes the pair of fingers **310**, which pulls the inlet **11** of the plastic bag **10** adhered to the elevating member **100** in the two opposite directions while moving in the directions that are away from each other, in the state inserted into the inlet **11** of the plastic bag **10**, a first driving part **320**, which allows the pair of fingers **310** to descend toward the plastic bag **10** so that the pair of fingers **310** are inserted into the inlet **11** of the plastic bag **10**, and a second driving part **330** which allows the pair of fingers **310** to move in the directions that are away from each other.
(41) The finger member **300** having such a structure may expand the inlet **11** of the plastic bag **10** in the two opposite directions, and may also prevent the inlet **11** of the plastic bag **10** from sagging.
(42) Movable Member
(43) The plastic bag opening device **1** according to the first embodiment of the present invention may further include a movable member **500** for secondarily expanding the inlet **11** of the plastic bag **10**.
(44) That is, as illustrated in FIG. **7**, the movable member **500** moves the elevating member **100** to be far away from the finger member **300** so that the inlet **11** of the plastic bag **10** is secondarily expanded by an area between the finger member **300** and the elevating member **100**. In other words, the first surface **12** of the plastic bag is adhered to the elevating member **100**, and the second surface **13** of the plastic bag **10** is supported by the finger member **400**. Thus, when the movable member moves the elevating member **100**, the inlet **11** of the plastic bag **10** may be expanded while the first surface **12** of the plastic bag **10** moves together with the elevating member **100**.
(45) Hereinafter, an opening method using the plastic bag opening device according to the first embodiment of the present invention will be described.
(46) [Plastic Bag Opening Method According to First Embodiment of Present Invention]
(47) As illustrated in FIG. **8**, a plastic bag opening method according to a first embodiment of present invention includes an elevating process, an opening process, a rotating process, a primarily expanding process, and a secondarily expanding process in order that an inlet **11** of a plastic bag **10** having a pocket shape is opened to be widened.
(48) The plastic bag opening method according to the first embodiment of present invention uses the plastic bag opening device according to the first embodiment of the present invention described above, and the plastic bag opening device **1** according to the first embodiment of the present invention includes an elevating member **100**, an air blower **200**, a rotating member **400**, a finger member **300**, and a movable member **500**.
(49) Elevating Process
(50) Referring to FIG. **2**, in the elevating process, the elevating member **100** is used to adhere a first surface **12** (a top surface in FIG. **1**) of a plastic bag **10** stacked at an uppermost side of stacked

plastic bags **10** and then to lift the plastic bag **10**.
(51) That is, the elevating member **100** adheres the first surface **12** of the plastic bag **10** through an adherence pad **110** and lifts the plastic bag **10** through an elevating part **120**. Here, the elevating member **100** lifts the plastic bag **10** up to a horizontal line (O) at which the air blower **200** is disposed.
(52) Here, the first surface **12** (the top surface in FIG. **1**) and the second surface **13** (a bottom surface in FIG. **1**), which are disposed at a side of an inlet **11** of the plastic bag **10**, may have an entirely close contact state or a partially close contact state by adhesion or static electricity.
(53) Opening Process
(54) Referring to FIG. **3**, in the opening process, the air blower **200** is used to inject air toward the inlet **11** of the plastic bag **10** adhered to the elevating member **100**. Then, the inlet **11** may be opened by pressure of the air injected from the air blower **200**. In particular, even when the first surface **12** and the second surface **13** that are disposed at the side of the inlet **11** of the plastic bag **10** are in close contact with each other, the first surface **12** and the second surface **13** may be easily separated from each other by the air pressure, and as a result, the inlet **11** of the plastic bag **10** may be opened to be widened.
(55) Furthermore, part of the air having flowed into the plastic bag **10** may be discharged to the outside through the inlet **11** of the plastic bag **10**, and the inlet **11** of the plastic bag **10** may be completely opened by the air discharged to the outside of the plastic bag **10**.
(56) In the opening process, the air blower **200** may inject the air toward the inlet **11** of the plastic bag **10** so that the air is injected downward at a set angle α with respect to the horizontal line. That is, the air blower **200** may inject the air toward the second surface **13** disposed at the side of the inlet **11** of the plastic bag **10** and accordingly, the second surface **13** of the plastic bag **10** may be easily separated from the first surface **12** of the plastic bag **10**. As a result, the inlet **11** may be effectively opened.
(57) The set angle α may be set to 3° to 10°, and preferably, may be set to 5° to 7°.
(58) When the inlet **11** of the plastic bag **10** is opened, the rotating process, the primarily expanding process, and the secondarily expanding process may be performed for expanding an area in an inlet **11** direction of the plastic bag **10** in order to easily accommodate articles.
(59) Rotating Process
(60) Referring to FIG. **4**, the rotating process is intended to dispose the inlet **11** of the plastic bag **10** to face upward, and when the opening process is completed, the rotating member **400** is used to rotate the elevating member **100** so that the inlet **11** of the plastic bag **10** adhered to the elevating member **100** is disposed to face upward.
(61) Primarily Expanding Process
(62) Referring to FIG. **6**, the primarily expanding process is intended to primarily expand the inlet **11** of the plastic bag **10** and prevent the inlet **11** from sagging, and when the rotating process is completed, a pair of fingers **310** provided in the finger member **300** are inserted into the inlet **11** of the plastic bag **10** adhered to the elevating member **100**, and the pair of fingers **310** are moved in directions that are away from each other. Then, the inlet **11** of the plastic bag **10** may be pulled by the pair of fingers **310** in two opposite directions to primarily expand the inlet **11** of the plastic bag **10**.
(63) Secondarily Expanding Process
(64) Referring to FIG. **7**, in the secondarily expanding process, the movable member **500** is used to move the elevating member **100** to be far away from the finger member **300** when the primarily expanding process is completed. Then, the first surface **12** of the plastic bag **10** moves together with the elevating member **100**, and as a result, the inlet **11** of the plastic bag **10** may be secondarily expanded by an area (or a distance) between the finger member **300** and the elevating member **100**.
(65) When the processes as described above are completed, the articles may be conveniently

inserted into the inlet **11** of the plastic bag **10** and packed.
(66) Hereinafter, another embodiment of the present invention will be described using the same reference symbol for the same element as the embodiment described above, and duplicate description will be omitted.
(67) [Plastic Bag Opening Device According to Second Embodiment of Present Invention]
(68) As illustrated in FIGS. **9** and **10**, a plastic bag opening device **1** according to a second embodiment of present invention includes an elevating member **100** and an air blower **200**. The air blower **200** includes an injection nozzle **210**, which injects air toward an inlet **11** of a plastic bag **10** so that the inlet **11** of the plastic bag **10** is opened to be widened, and a frame **220** to which the injection nozzle **210** is coupled.
(69) Here, the injection nozzle **210** may be coupled to the frame **220** so that an angle is adjustable up and down or left and right.
(70) In one example, referring to FIG. **10**, a spherical coupling part **211** is formed in the injection nozzle **210**, and a spherical coupling groove **221** to which the spherical coupling part **211** is coupled is formed in the frame **220**. That is, the spherical coupling part **211** may rotate up and down or left and right within the spherical coupling groove **221**, and the injection nozzle **210** may operate with the spherical coupling part **211** to adjust the angle up and down or left and right.
(71) Thus, in the plastic bag opening device according to the second embodiment of present invention, the angle of the injection nozzle **210** may be adjusted up and down or left and right to match the inlet **11** of the plastic bag **10** adhered to the elevating member **100**. As a result, the air injected from the injection nozzle **210** may be accurately injected into the inlet **11** of the plastic bag **10**.
(72) The scope of the present invention is defined by the appended claims rather than the foregoing description. Various modifications made within the meaning of an equivalent of the claims of the invention and within the claims are to be regarded to be in the scope of the present invention.

Claims

1. A plastic bag opening device configured to open an inlet of a plastic bag having a pocket shape, the plastic bag opening device comprising: a lifter configured to lift the plastic bag in a state in which a first surface of the plastic bag is adhered to the lifter; an air blower configured to inject air toward the inlet of the plastic bag adhered to the lifter so that the inlet of the plastic bag is opened; a pair of fingers configured to be inserted into the inlet of the plastic bag adhered to the lifter to pull and expand the inlet of the plastic bag in opposite directions; and a movable member configured to move the lifter away from the pair of fingers such that the inlet of the plastic bag is expanded between the pair of fingers and the lifter.
2. The plastic bag opening device of claim 1, wherein the lifter comprises: an adherence pad configured to adhere to the first surface of the plastic bag; and an elevating part configured to cause the adherence pad to ascend so as to lift the plastic bag adhered to the adherence pad up to an imaginary horizontal line at which the air blower is disposed.
3. The plastic bag opening device of claim 2, wherein the air blower comprises: an injection nozzle configured to inject the air toward the inlet of the plastic bag so that the inlet of the plastic bag is opened, the injection nozzle having an air injection opening configured to be inclined downward at a set angle with respect to the imaginary horizontal line; and a frame to which the injection nozzle is coupled.
4. The plastic bag opening device of claim 3, wherein the set angle is set to 3° to 10°.
5. The plastic bag opening device of claim 2, wherein the air blower comprises: a frame; and an injection nozzle configured to inject the air toward the inlet of the plastic bag so that the inlet of the plastic bag is opened, the injection nozzle being coupled to the frame so that an angle of the injection nozzle with respect to the frame is adjustable up and down or left and right.

6. The plastic bag opening device of claim 1, further comprising a rotator configured to rotate the lifter so that the inlet of the plastic bag adhered to the lifter faces upward.
7. A plastic bag opening method that opens an inlet of a plastic bag having a pocket shape, the plastic bag opening method comprising: an elevating process of adhering a first surface of the plastic bag to a lifter and then lifting the plastic bag using the lifter; an opening process of opening the inlet of the plastic bag using an air blower to inject air toward the inlet of the plastic bag adhered to the lifter so that the air injected from the air blower flows into the plastic bag through the inlet; after the rotating process, a primary expanding process of pulling and expanding the inlet of the plastic bag in opposite directions by inserting a pair of fingers into the inlet of the plastic bag adhered to the lifter and moving the pair of fingers in directions that are away from each other; and after the primary expanding process, a secondary expanding process of expanding the inlet of the plastic bag between the pair of fingers and the lifter using a movable member to move the lifter away from the pair of fingers.
8. The plastic bag opening method of claim 7, wherein, in the elevating process, the lifter lifts the plastic bag up to an imaginary horizontal line at which the air blower is disposed, and wherein, in the opening process, the air blower injects the air toward the inlet of the plastic bag so that the air is injected downward at a set angle with respect to the imaginary horizontal line.
9. The plastic bag opening method of claim 8, wherein the set angle is set to 3° to 10°.
10. The plastic bag opening method of claim 7, further comprising, after the opening process, a rotating process of rotating the lifter using a rotator so that the inlet of the plastic bag adhered to the lifter is disposed to face upward.